

NETWORK PROGRAMMING

CNS 2205

CAT 1 – 15 Oct 2025

Attempt All Questions

Time: 1 Hour

Answer All Questions [25 Marks]

Question 1 (10 Marks):

- a. Give the name of a software that can provide each of the following server applications on a Linux system. Database server, DHCP server, Proxy server. [3 Marks]
- b. Explain why authentication of users accessing the system using SSH should be accomplished by Public Keys instead of Usernames and Password [3 Marks]
- c. Using a diagram with an SSH client and server, describe how SSH public key authentication works. [4 Marks]

Question 2 (8 Marks):

- a. On a Linux system
 - i. Give the full path and name of files used to configure the hostname [1 Mark]
 - ii. Give the full path and name of files used to configure the time zone. [1 Mark]
 - iii. Give the command that would set the hostname to ComputerNet [1 Mark]
 - iv. Give the full path to the file that an administrator can be used to map IP addresses and computer names if a DNS server was not accessible. [1 Mark]
- b. Name two Linux files that are used for holding login credentials for system users. [2 Marks]
- c. State the name and describe the use of a utility that be used to copy files between Linux computers. [2 Marks]

Question 3 (7 Marks):

- a. Write a cron job that uses rsync to back up the /home/user/Documents directory every day at 11:30 p.m to the /home/Backups directory on a computer whose IP address is 10.20.113.44. [4 Marks]
- b. Set the permissions of a file to rwxr-xr-- using both numeric and symbolic methods. [2 Marks]
- c. Give the command for counting the number of lines in the file passwd.txt [1 Mark]

Q1

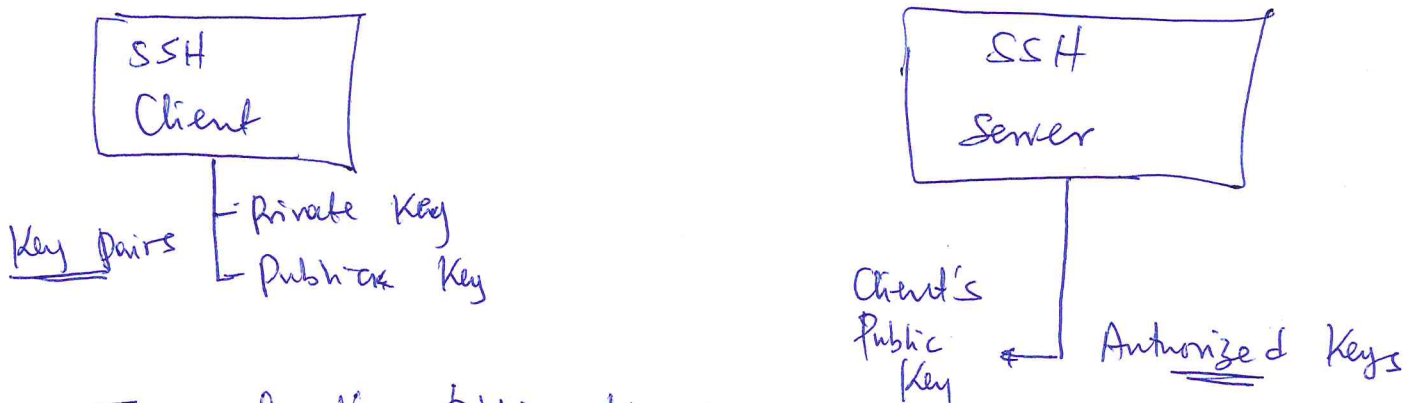
(a) Database Server: MySQL, MariaDB, MongoDB, MS SQL

DHCP server: ISC-dhcp-server, dnsmasq

Proxy server: Squid, HAProxy, Apache Proxy

- (b) - PKA is safe from stolen ~~password~~ credentials as long as private key is secured
- Passwords can be guessed
 - PKA uses strong credentials → safe from weak passwords
 - PKA is safe from MITM that can intercept passwords
 - PKA is safe from shoulder surfing that can steal passwords.

(c)



- The client's public key is copied onto the server
- When client connects, it presents public key.
- L server creates challenge and sends to client to solve using matching private key
 - Client presents solution and gets authenticated.

Q2

(a) (i) hostname — /etc/hostname

(ii) Timezone — /etc/timezone

(iii) timezonectl set-hostname ~~the hostname~~ ComputerNet

(iv) /etc/hosts

(b) /etc/passwd

/etc/shadow

(c) — rsync — Rsync takes the location of source and destination of files including IP address of remote computer.

→ ~~scf~~ — It uses ssh for ~~scf~~ authentication and secure transport

— scp — Takes location of source and dest including IP of remote computer

↳ Also use ssh for auth and ~~for~~ secure transport

Q3

(a)	<u>mm</u>	<u>hh</u>	<u>dd</u>	<u>month</u>	<u>day</u>	<u>user</u>	<u>Command</u>
	30	23	*	*	*	root	/usr/sbin/rsync -avz
						10.20.113.44:	/home/user/documents
						/home/Backups/	

← Cont...

← Cont...

(b) chmod 754 file

chmod u+rwx, g+rx, O+r

(c) wc -l passwd.txt